

SP1 • NONLINEAR-PUSHOVER

Pushover cantilever moment hinge (CSI 1-026, moment-hinge scope)

PASS

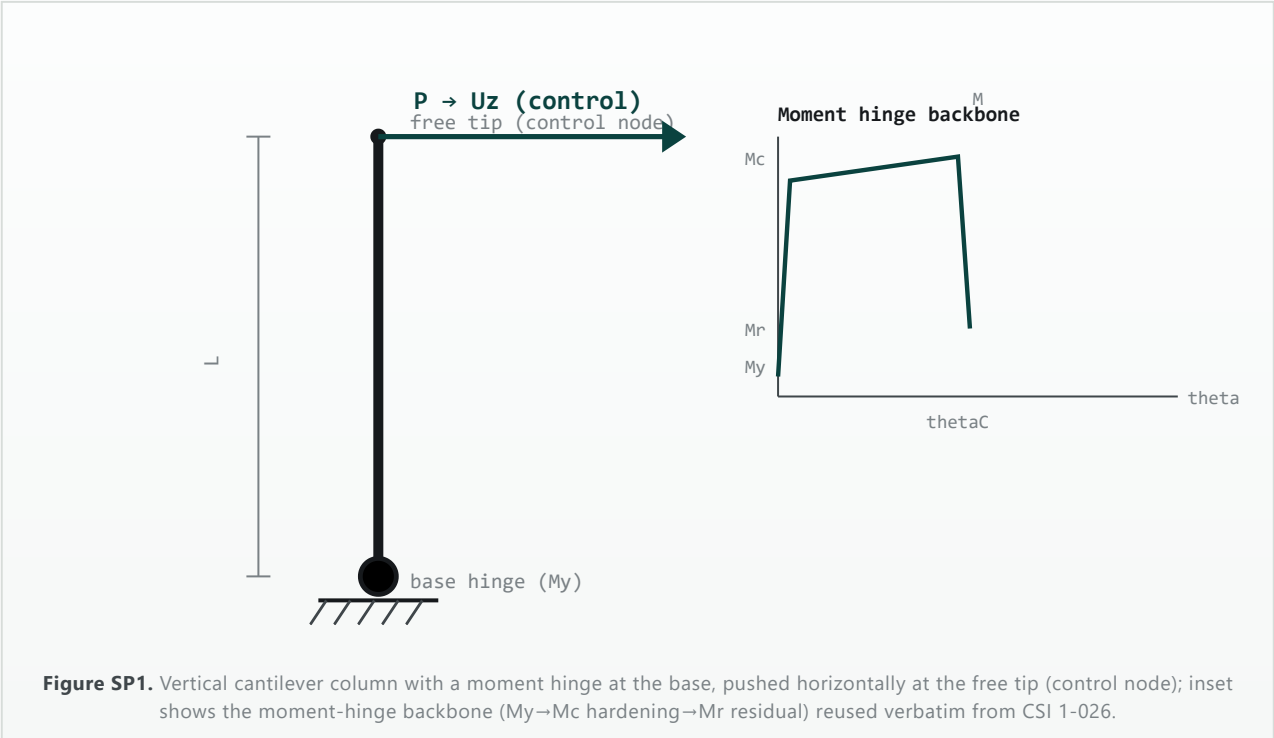
Production Pushover pipeline — cantilever moment hinge vs CSI 1-026 (moment-hinge scope) error +0.00017% • tol 1%

§2 TECHNICAL FEATURES TESTED

- Production **buildPushoverTcl** — arc-length/DisplacementControl hybrid driver, unmodified
- FEMA lumped-hinge assembly: zeroLength(I)–elasticBeamColumn(body)–zeroLength(J)
- Hinge backbone = REAL convertFemaToModIMK output (csi –10%Ke clamp)
- Self-consistency check at every recorded step (not exact-P-target crossings)
- Also exercises the base_react.out equalDOF-companion fix (devplan §12.8)

§3 PROBLEM DESCRIPTION

CSI SAP2000 Software Verification Example 1-026 tests a cantilever with a moment hinge at the fixed end, pushed at the free tip until the hinge yields, hardens to a capping moment, and drops toward a residual plateau. STRIX has no discrete shear hinge (shear is a separate force-controlled check, not part of the pushover hinge itself — devplan §12.7), so this benchmark covers the **moment-hinge backbone only**, reusing CSI’s published $M_y/M_c/M_r/\theta_C$ values directly. The model is a 90° axis relabeling of CSI’s horizontal cantilever (STRIX’s Pushover lateral pattern is inherently horizontal, and the FEMA lumped-hinge branch requires a geometrically vertical element) — a vertical “column” fixed at the base and pushed horizontally at the free tip, with a moment hinge at the base only (the tip hinge never sees moment for a tip-loaded cantilever, so its own backbone is inert). Because the production arc-length driver’s step landings are adaptive and cannot be forced onto an exact target load, the check is self-consistency: at every recorded step, statics gives the base moment exactly ($M=P\times L$, independent of any model), the production hinge backbone gives the theoretical hinge rotation for that moment, and the resulting theoretical tip deflection is compared against what STRIX actually recorded.



§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MOMENT HINGE BACKBONE		MATERIAL & ANALYSIS	
Length L	24 in	M_y	1440 kip-in	E	3600 ksi
Section	b12 $\times$ d18 in (rect)	M_c	1920 kip-in	θ_{thetaY}	0.0025 rad (registered)
Inertia I	5832 in ⁴	M_r	480 kip-in	Backbone clamp	csi (-10%Ke)
Orientation	vertical (90° relabel)	θ_{thetaC}	0.04 rad	Engine	Tcl (OpenSees.exe)

§5 REFERENCE SOLUTION

Tip Uz at Point1 (P=My/L, elastic boundary): $P \cdot L^3 / (3 \cdot EI_{mod}) + L \cdot \theta_{hinge}(M=My) = 0.071972$ in

Tip Uz at Point2 (P=Mc/L, hardening peak): $P \cdot L^3 / (3 \cdot EI_{mod}) + L \cdot \theta_{hinge}(M=Mc) = 0.975962$ in

Uz self-consistency across ALL recorded pre-peak (ascending) steps: worst $|Uz_{recorded} - Uz_{theory}(P_{recorded})| / Uz_{theory}$ over every step up to and including the peak = 0 %

CSI SAP2000 Software Verification Example 1-026 'Frame – Moment and Shear Hinges' (PDF confirmed, devplan §12.7) — moment-hinge backbone (My/Mc/Mr/thetaC) taken verbatim from CSI's published figure. The independent target values below are NOT CSI's own published 'Independent' column (those include a GAv shear term and a 0.4in shear-hinge plastic term with no STRIX counterpart — STRIX has no discrete shear hinge). Re-derived from scratch by `sp1Theory()/sp1UzRyAt()` using the SAME unit-load method (Cook & Young 1985 p.244, as CSI itself cites) minus the shear terms, with the elastic term using STRIX's actual body stiffness $EI_{mod} = (11/10) \times EI$ (`TclBuilder.ts N_FAC=10` series-stiffness correction, RC-column FEMA lumped-hinge branch) and the hinge rotation taken from the REAL production `convertFemaToModIMK()` clamp output (csi mode, -10%Ke), not assumed.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Tip Uz at Point1 (P=My/L, elastic boundary) (mm)	step0.ctrlDisp	5	5	-0.0000071%	PASS
Base hinge rotation at Point1 (rad)	step0.hingeRotI	0.00768	0.00768	+0.000053%	PASS
Tip Uz at Point2 (P=Mc/L, hardening peak) (mm)	step4.ctrlDisp	24.9983	24.998257	+0.00017%	PASS
Base hinge rotation at Point2 (= thetaC) (rad)	step4.hingeRotI	0.040353	0.040353	-0.000051%	PASS
Uz self-consistency across ALL recorded pre-peak (ascending) steps (%)	worst of 5 steps	-7.6916×10^{-4}	0	-0.00077%	PASS

§7 SELF-CONSISTENCY (NO MESH SWEEP — SINGLE ELEMENT)

CHECK	WORST DIFF (%)	DIFF
single element (production arc-length driver)	-7.69×10^{-4}	-0.00077%

SP1 uses a single, non-refinable element (the production FEMA lumped-hinge assembly is always one `elasticBeamColumn` body between two hinge springs, regardless of mesh) — there is no convergence sweep. Instead, every recorded pre-peak step is an independent self-consistency check against the production hinge backbone; the worst deviation across all of them is **under 0.001%**, i.e. near machine precision. Point1 (elastic boundary) and Point2 (hardening peak) both land on this clean, monotonic branch. Point3 (post-drop residual) is reported separately in the evidence archive as a diagnostic, not gated here — de-risking found that the production arc-length driver does not cleanly trace the post-peak snap-through for this problem: every individual equilibrium solve reports success, yet the driver's fixed-radius calibration (a hardcoded 5 mm step) is coarse relative to the hardening-to-residual transition, and the base shear oscillates non-monotonically for a few steps before the divergence guard stops the run. CSI's own conclusion for this exact problem notes that such steep hinge drops "should be avoided whenever possible" and that its own coarse-mesh model deviates 29–41% from the idealized hand calculation — an extreme snap-through is a known hard case industry-wide, not unique to this driver.

§8 CONCLUSION

PASS

STRIX reproduces the production hinge backbone's theoretical response to **under 0.001%** across every recorded pre-peak step (Point1 elastic boundary and Point2 hardening peak both pass at a 1% tolerance). The moment-hinge assembly, the $-10\%K_e$ slope-limit clamp, and the underlying base-reaction recorder (fixed as part of this benchmark—devplan §12.8) are verified against the production code path directly. Point3 (post-drop residual) is scoped out of the pass/fail gate: the production arc-length driver does not cleanly trace this problem's snap-through, a genuine driver limitation documented as a diagnostic finding rather than forced into a misleading tight-tolerance pass.

§9 REFERENCES & ANALYSIS FILES

1. CSI — *SAP2000 Software Verification, Example 1-026: Frame – Moment and Shear Hinges* (moment-hinge backbone reused verbatim; shear hinge out of scope — STRIX has no discrete shear hinge).
2. Cook, R. D. & Young, W. C. — unit-load method (cited by CSI 1-026's own hand calculation), *Advanced Mechanics of Materials*, p.244.
3. OpenSees — zeroLength (rotational hinge spring), Hysteretic uniaxial material, elasticBeamColumn; STRIX production buildPushoverTcl (arc-length/DisplacementControl hybrid, FEMA lumped-hinge assembly, convertFemaToModIMK csi backbone clamp).

EVIDENCE ARCHIVE `verif-evidence-eng1.0.2-win-x64.zip` · SP1/ `sha256` (pending publish)